A Win Probability Strategy for the Power Play in Mixed Doubles Curling

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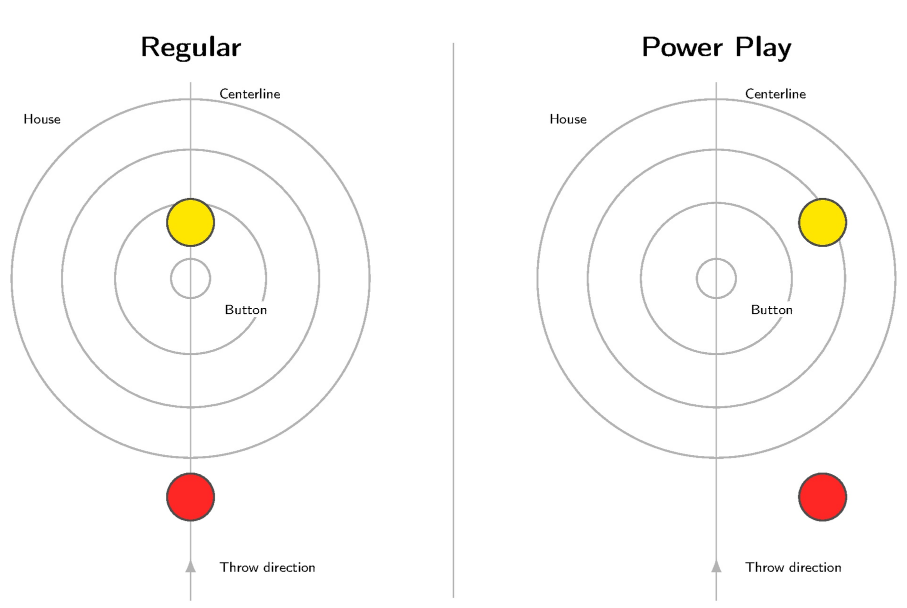
The Question

In mixed doubles curling, each team gets **one Power Play per game**, which shifts pre-positioned stones to open the scoring lane. Since teams only have one to use, the big question is:

When should you deploy it?

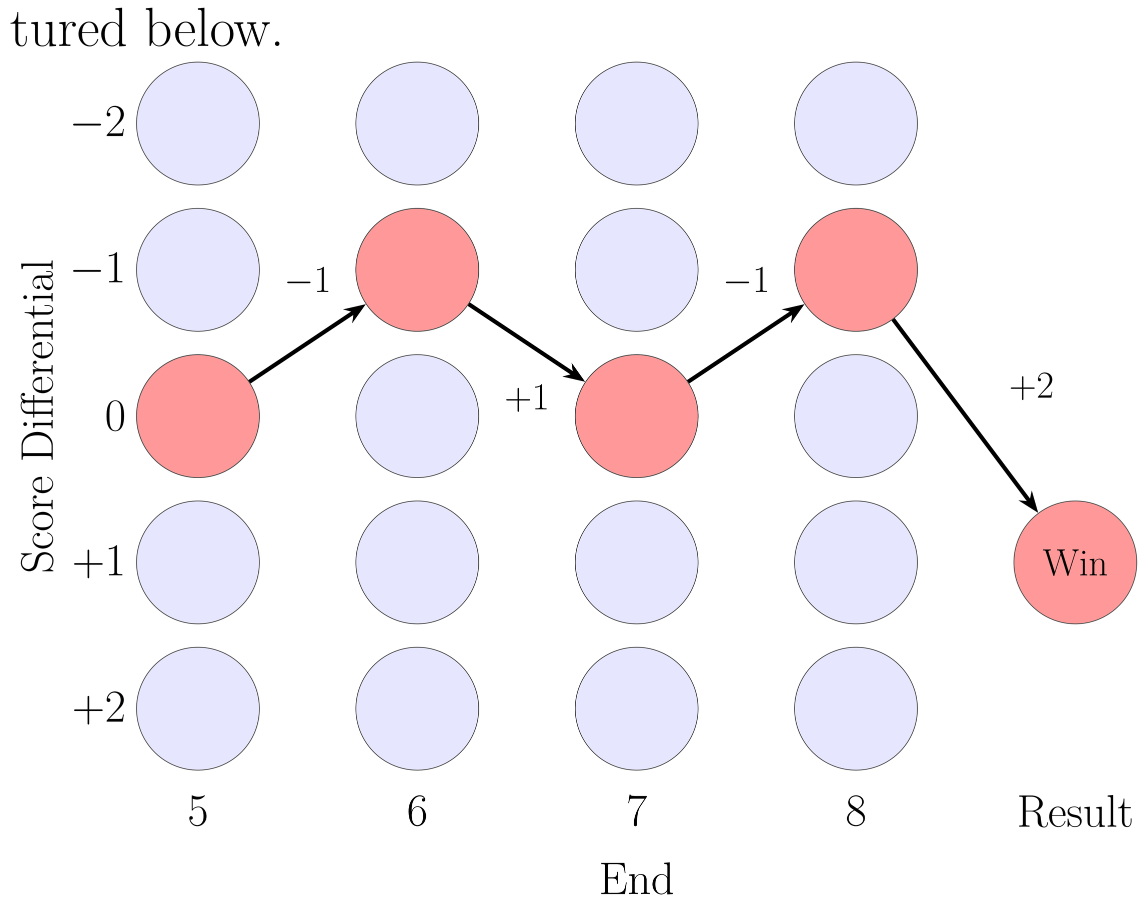
The Power Play



Regular setup (left) vs. Power Play (right).

Modeling Approach

Reasoning based on expected points can conclude "use PP earlier" because PP adds points in any end. But a point in end 3 is **not** worth the same as a point in end 8: optimizing EP ignores the **option value** of saving a scarce resource for a higher-leverage moment. Instead, we model the match as a **finite-horizon MDP** [1] with states $s_e = (e, d_e)$, as pictured below.



Mathematical Solution

1. Transitions: A multi-class XGBoost model conditions on hammer, PP availability, and relative team strength (Elo) to estimate the end-outcome distribution $P(\Delta_e \mid s_e, a)$ from **5,274** end-states, **344** games [2].

2. Solve: Backward induction yields the optimal stopping rule:

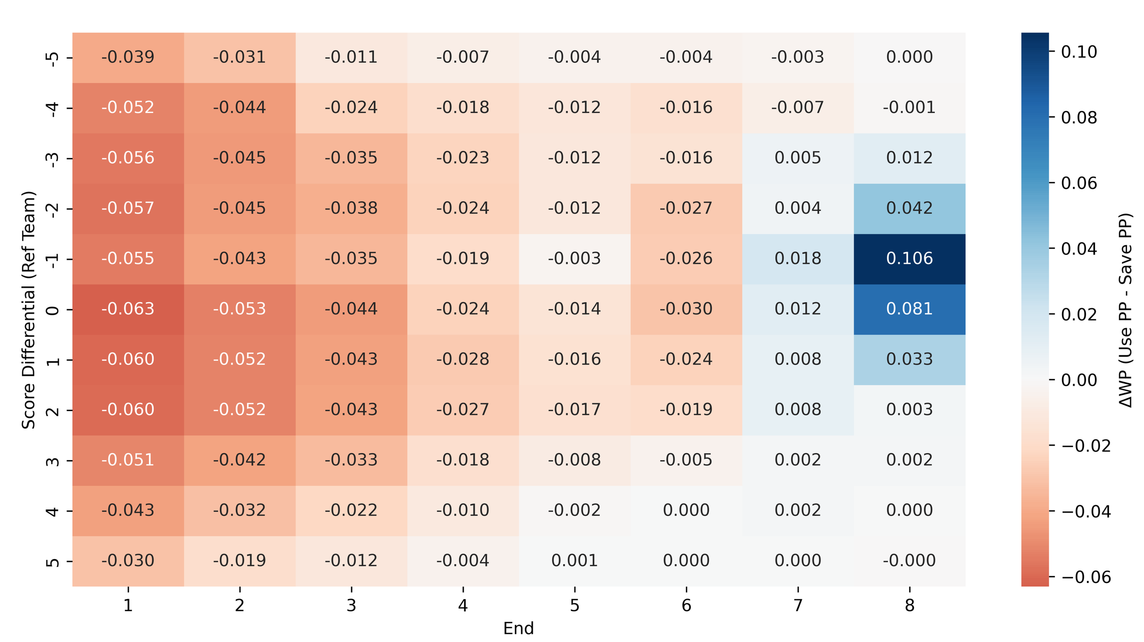
$$\Delta\text{WP} = V(s_e \mid \text{use}) - V(s_e \mid \text{save}) > 0$$

i.e., use PP if it increases the estimated win probability relative to saving for later.

Key Result

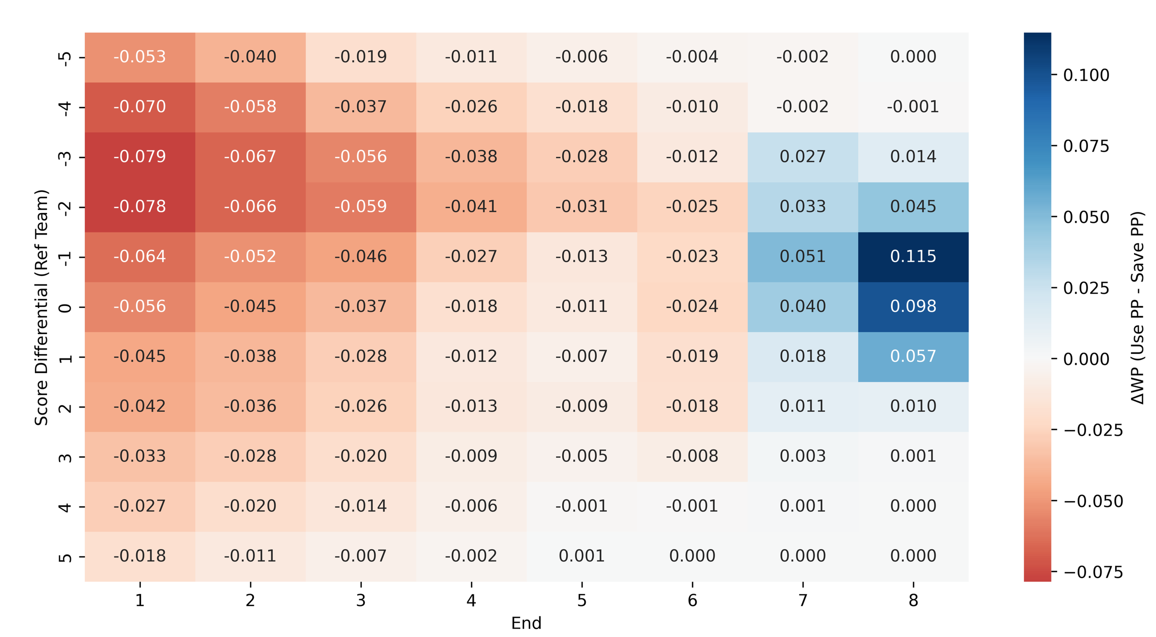
End 8, down 1: PP adds **+10.6%** win probability. Tied: **+8.1%**. Earlier in the game, the **option value of saving** almost always dominates.

Optimal Policy: Opp PP Avail.



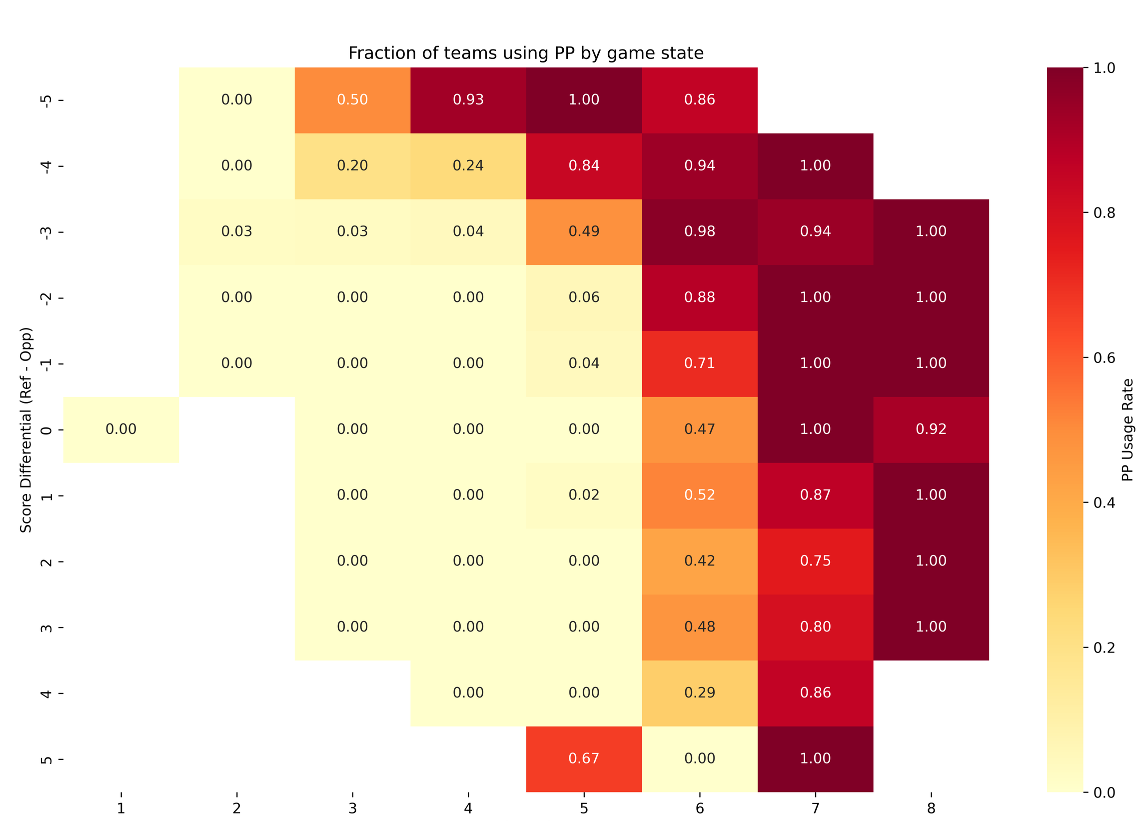
ΔWP by end and score. Blue = use now; red = save.

Optimal Policy: Opp PP Used



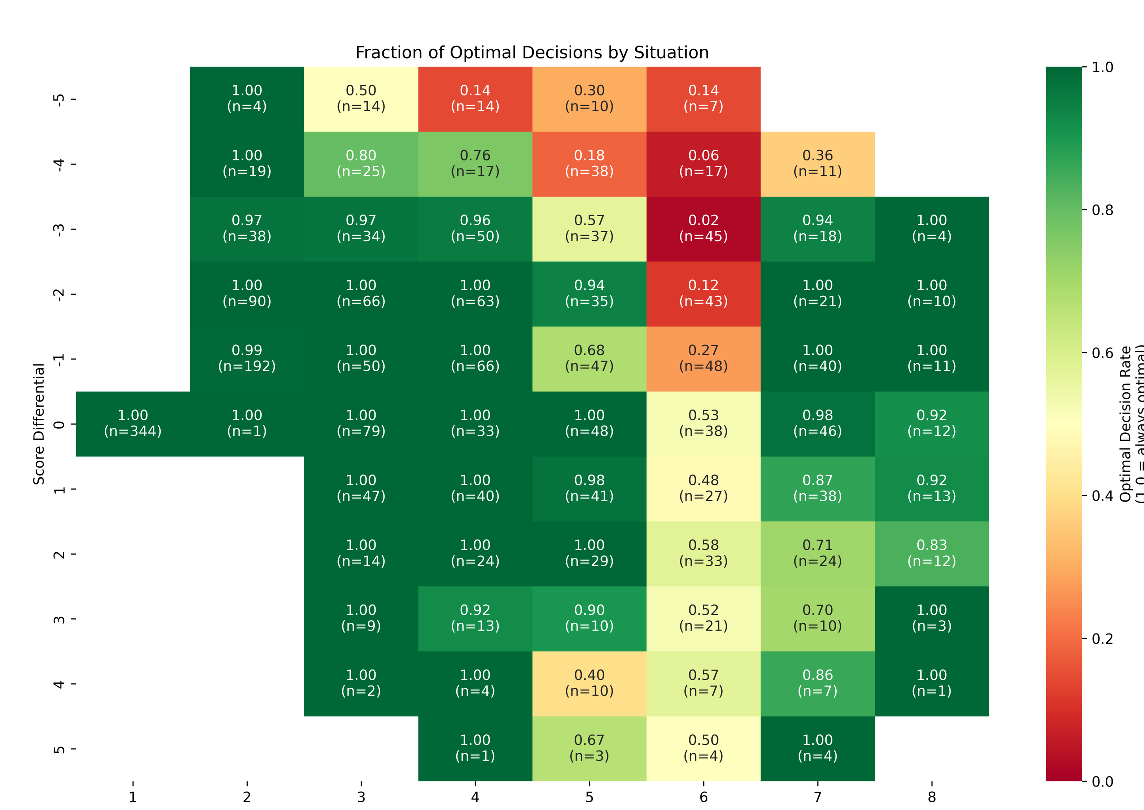
PP is worth *more* when the opponent has already used theirs.

Observed PP Usage



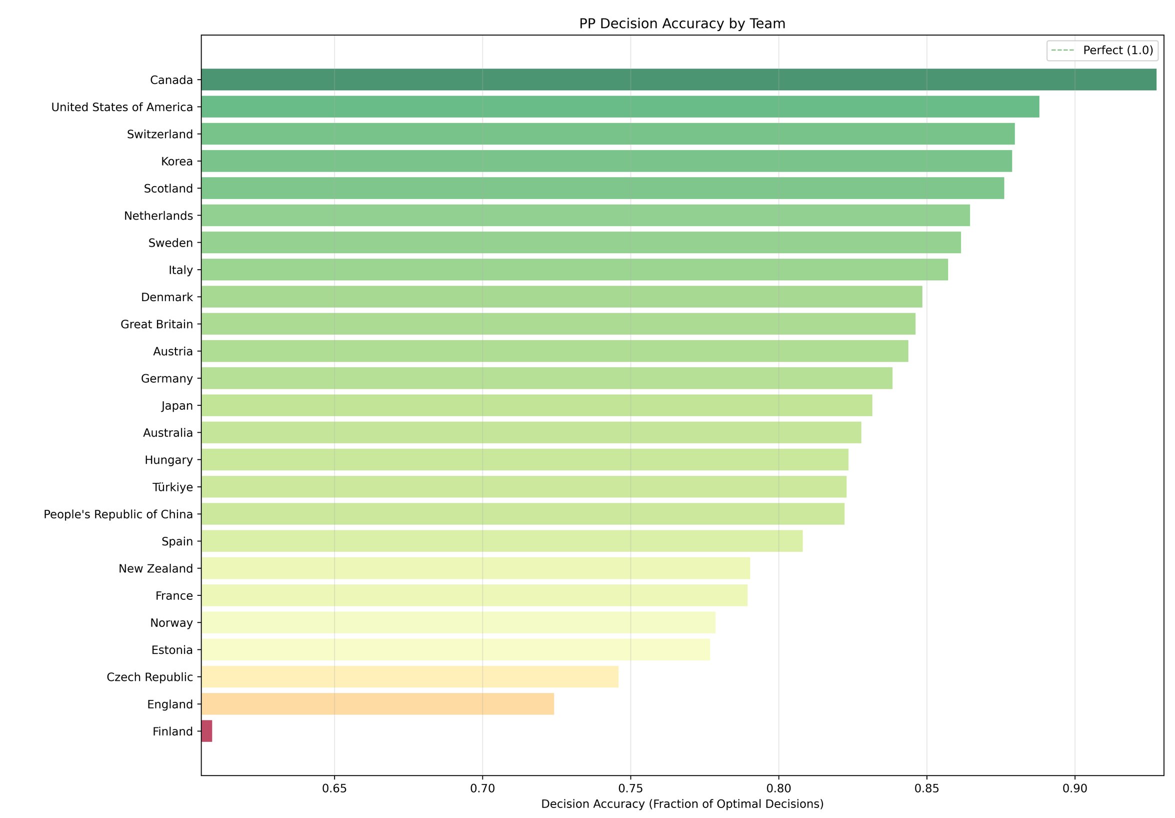
When teams actually deploy PP. Most used in ends 5–8.

Decision Accuracy vs. Optimal

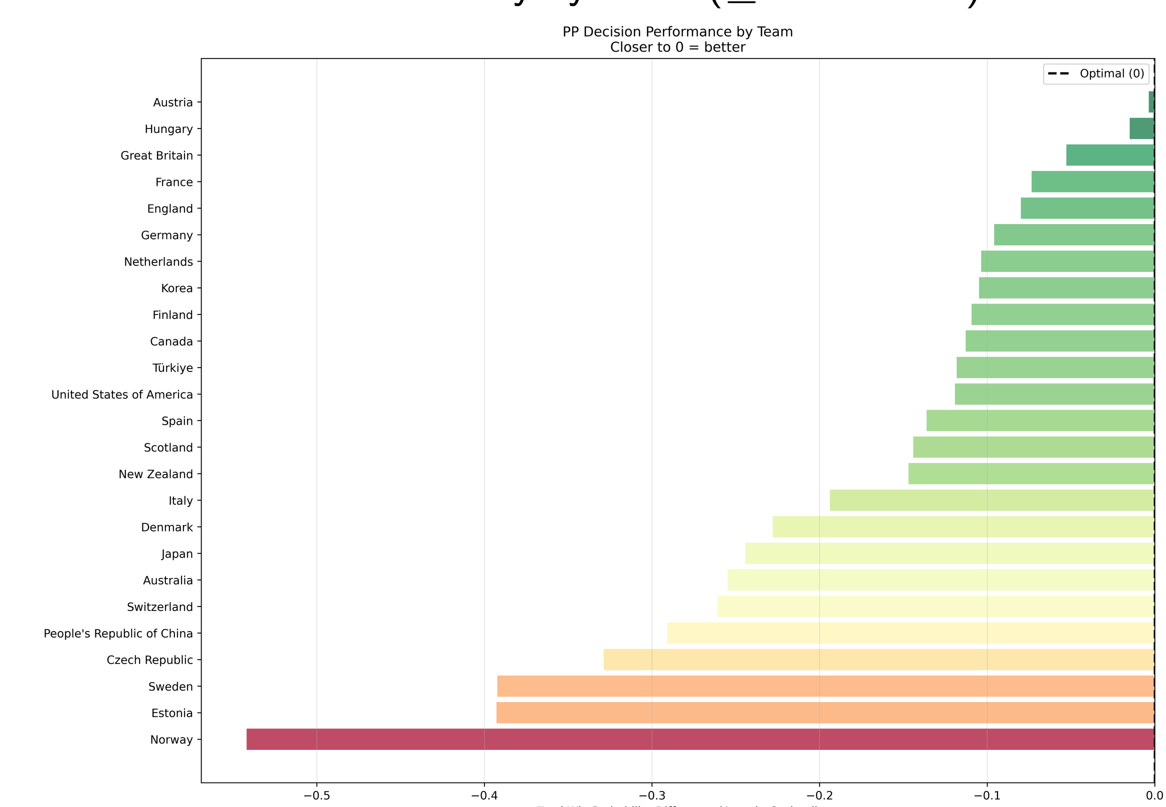


Biggest departures: early use in end 6 (up to 1.8% WP loss).

Team-Level Performance



Decision accuracy by team (≥ 5 decisions).



Total WP impact of PP timing decisions by team.

Practical Guidance

Ends 1–6: Default to **saving**: option value is large.

Ends 7–8: **Use** in almost all game states.

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References

[1] Kostuk et al., "Modelling curling as a Markov process," *EJOR*, 2001.

[2] CSAS 2026 Data Challenge, `statds.org/events/csas2026`